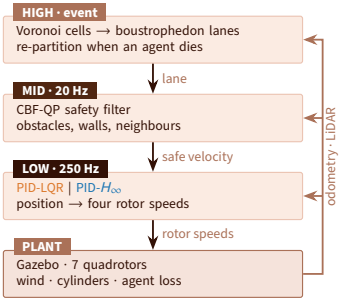
Fault-Tolerant Swarm Quadrotor Control for Non-Convex Geodetic Mapping

Izma Alhazmi Herdian[1,2] **Sulthan Naufal Affan**[1]
Estiyanti Ekawati[1,2] **Faqihza Mukhlish**[1,2]

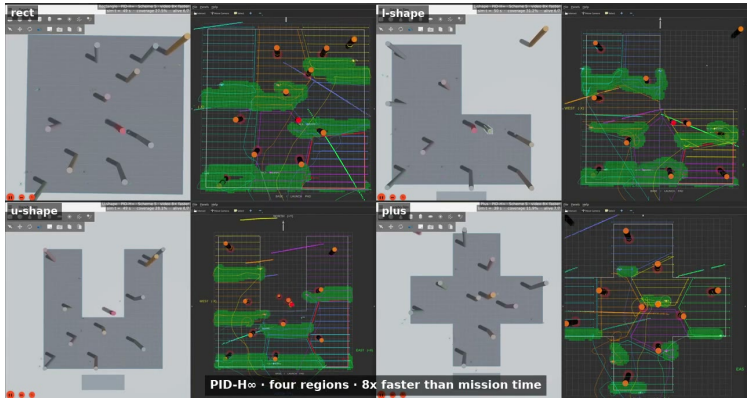
[1]Engineering Physics Department, Institut Teknologi Bandung
[2]Center for Instrumentation Technology and Automation, ITB

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github.com/izmaherdian/swarm-quadrotor-mapping

Seven quadrotors, four regions, every stressor at once



PID-H_∞ on rect, L, U and plus · 8× faster than mission time · Gazebo left, RViz map right · Click the frame to play

The question is not whether a swarm is faster, but whether it still finishes

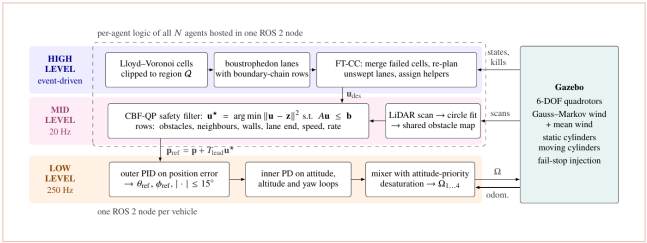
- Non-convex region — notches and arms break a simple back-and-forth sweep
- Wind — mean 3.0 m/s, gusts up to 12.0 m/s
- Obstacles not known in advance, found in flight by LiDAR
- Two of seven quadrotors lost mid-mission

> **ALL FOUR AT ONCE**
> Not one at a time — the swarm has to keep mapping while every one of these is happening.

Four contributions

- Lanes that follow each cell's shape, so tapered corners get swept
- Lane reallocation when an agent is lost — merge, re-plan, hand over
- A safety filter whose speed limit comes from the real tilt limit
- PID-LQR vs PID-H_∞ on an identical structure — only the gains differ

Three layers, each one handing a target to the next



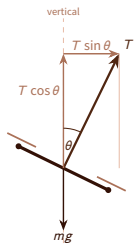
HIGH · event-driven
in region polygon, agent poses, agent losses
out a lane to follow

MID · 20 Hz
in desired velocity, LiDAR map, neighbours
out a safe velocity

LOW · 250 Hz
in position target, state
out four rotor speeds

Low level: one tilt limit, one structure, two controllers

- Only tilted thrust moves a quadrotor sideways, so capping tilt at 15° caps acceleration at $2.631\,\mathrm{m/s^2}$
- PID-LQR and PID-H_∞ are designed on the same model and run as the same PID
- They differ only in their objective — the model, or the worst disturbance



PASSED UPWARD: FOUR NUMBERS, FROM THE GAINS

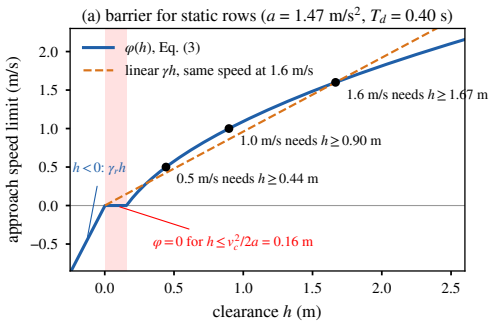
k_v $3.867\,\mathrm{s^{-1}}$ · $a_{\max}$ $2.631\,\mathrm{m/s^2}$
T_{lead} $0.323\,\mathrm{s}$ · v_c $0.680\,\mathrm{m/s}$

Mid level: a speed limit the drone can actually obey

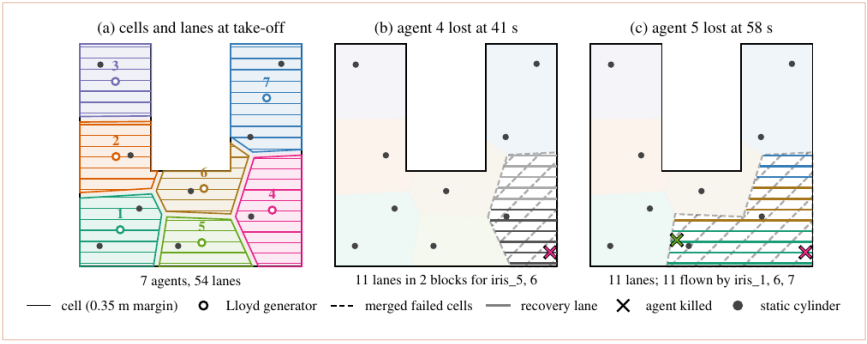
- Tilt limit + 0.4 s reaction delay: the fastest approach that can still stop, at every distance
- Each obstacle, wall or neighbour limits only the *approach* speed — sideways stays free
- A small QP picks the safe velocity closest to the plan, 20 times per second

WHY NOT A SIMPLE LINEAR RULE

It allows approach where the drone can no longer stop (pink band). Ours has no free parameter.



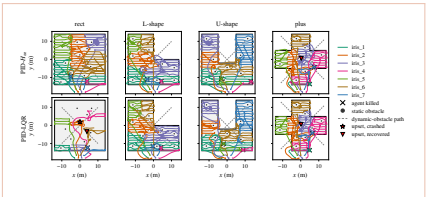
High level: cells, lanes, and re-planning when an agent is lost



(a) 7 agents, 54 lanes · (b) agent 4 lost at 41 s · (c) agent 5 lost at 58 s

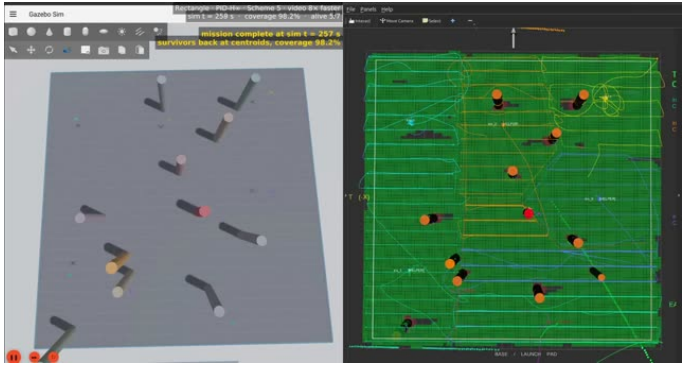
Every stressor applied at once, to seven quadrotors

- Seven 3DR Iris in Gazebo + ROS 2 · cruise 2.0 m · sweep 1.6 m/s
- Turbulent wind, peaks 12.0 m/s
- Unknown static cylinders, plus two moving ones
- Two agents lost, at 18% and 32% coverage
- Four regions × two controllers = eight missions



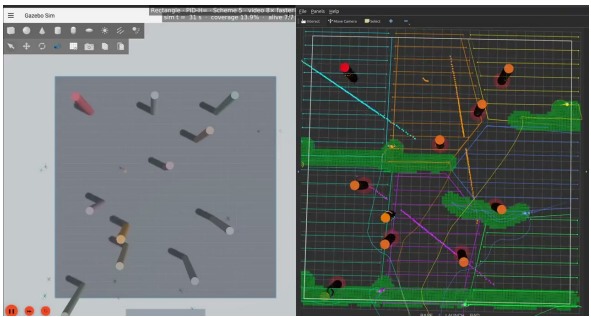
Flown trajectories — PID-H_∞ (top), PID-LQR (bottom)

A full mission: partition, transit, sweep, two losses, return



rect, PID-H_∞ · 8× wall clock ≈ real mission time · 98.2% in 257 s, agents 4, 7 lost · Click the frame to play

The swarm never pauses — survivors keep sweeping through a reallocation



rect, PID-H_∞ · 25 s of mission time around the first loss · Click the frame to play

- Agent 4 lost at 18% coverage
- Its cell's coverage is reset
- The cell is merged and re-planned
- Helpers take over its lanes

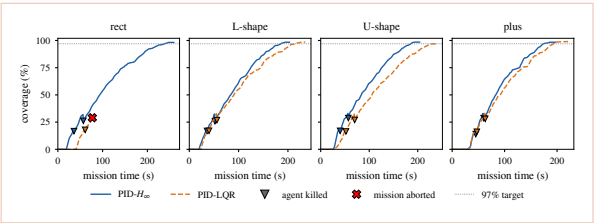
Seven of eight missions finished; PID-H_∞ finished every region

Region	Controller	Status	Cov. (%)	T (s)	Lat. RMS (cm)	Effort
rect	PID-H_∞	done	98.2	257	7.2	0.054
rect	PID-LQR	**stopped @ 78.8 s**	n/a	n/a	11.7	0.024
L-shape	PID-H_∞	done	98.4	201	8.4	0.058
L-shape	PID-LQR	done	98.5	233	12.2	0.010
U-shape	PID-H_∞	done	98.5	202	8.3	0.044
U-shape	PID-LQR	done	96.9	237	11.4	0.011
plus	PID-H_∞	done	98.4	195	9.7	0.053
plus	PID-LQR	done	99.0	219	11.5	0.014

REPORTED, NOT EXCLUDED

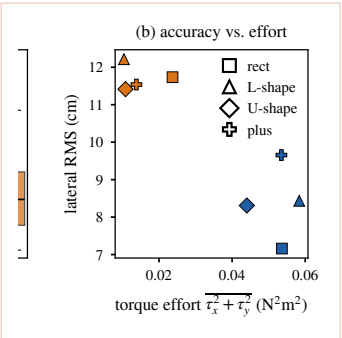
rect / PID-LQR: one agent struck a cylinder and the run was stopped — counted as a failure.

Coverage drops at every loss, then recovers



- Coverage inside a lost cell is reset and swept again
- The swarm recovers after each loss
- PID-H_∞ (solid) reaches the 97% target sooner in every region both completed

PID-H_∞ holds the lane tighter, at a torque cost



Lateral RMS against torque effort · marker shape is the region

- Lateral error 16–39% lower with PID-H_∞
- Mostly a removed steady offset — its integral action is 3.9× stronger
- Torque effort 3.9–5.6× higher; motor saturation below 0.2%

NEITHER DESIGN WINS OUTRIGHT

PID-H_∞ is tighter on the lane; PID-LQR is gentler on the airframe.

Seven of eight, under every stressor at once

- 96.9–99.0% coverage while losing two of seven quadrotors
- Where both finished, PID-H_∞ held lanes 16–39% tighter and finished 11–15% sooner — at several times the torque
- One PID-LQR mission struck a cylinder, and is reported as such
- Scope: simulation, one run per configuration
- Next: more wind conditions, distributed planning, flight tests

Thank you

Questions welcome

Izma Alhazmi Herdian · izmaherdian@gmail.com
Engineering Physics, Institut Teknologi Bandung

Code MIT, media CC BY 4.0 · all eight mission recordings are public
Acknowledgment: CITA ITB covered registration and publication fees



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